A Systematic Literature Review on Test Case Prioritization in Combinatorial Testing

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ABSTRACT

In recent years, regression testing is widely used and performed in industries during the maintenance phase of software system development. The increases of software regression test cases resulted in a higher complexity of the software system, due to the rapid change in the software development. Thus, many researchers had proposed the test cases prioritisation technique to overcome the raised issues. However, most of the applications needed the combinatorial testing due to the enormous size of the software inputs. The problem arises when the small t-way values were selected in combinatorial testing, which resulted in the ineffectiveness of the fault detection ability. This work presents the results of a systematic literature review on the test case prioritisation and combinatorial testing that report the main approach, problems, and evaluation metric that were being used from existing studies. This paper aims to identify the aspects of the test case prioritisation and combinatorial testing that have been studied before, provide a basis for the improvement, and evaluate the current trends of this research area. Apart from the recent works that were published in the last ten years, 20 papers were selected for further study in the systematic literature review. The results show that this area of study is open for improvement especially to efficiently generate the test cases, which minimise the number of test cases generated, and effectively prioritise the test cases that increase the fault detection ability in software testing. However, the use of both techniques still needs more enhancements for the existing proposed approach that could result in the increasing of the fault detection ability.

CCS CONCEPTS

• **Software and its engineering**; • **Software testing and debugging**;

KEYWORDS

Software testing, Test case prioritization, Combinatorial testing, Systematic literature review

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1 INTRODUCTION

Software Testing is a process, or a series of processes, designed to make sure computer code does what it was designed to do and to ensure that the software does not do anything unintended [1]. To ensure that the software quality is up to the standard of the customer expectation, the modified software needs to undergo regression testing [2]. However, to perform regression testing in software testing, it will consume a huge amount of cost [3]. The goal of the regression testing is to focus on exploring the bugs or errors that possibly occur due to the change [4]. For the past few years, several techniques had being used to solve this problem, such as test case minimisation (TCM), test case selection (TCS) and test case prioritisation (TCP) but the TCP technique is on the rise of study due to its ability to perform well in detecting fault as early as possible [5]. Thus, different approaches have been proposed in the past few years to support this technique.

However, in most of the applications where Combinatorial testing (CT) is also needed, it has become impractical due to the enormous size of the software inputs such as the parameters and the value dependencies which consume the time taken for testing the test suite [6]. CT were explored with the goal to minimise the amount of test case generation from the large scale software [7]. The used of t letter in represented as the interaction strength [8]. Nowadays, researcher had proposed various approaches concerning this technique including repeated small-strength t-way testing, using the increment strength of the t-way testing in the regression testing and various other optimisation algorithms. Despite the fact that there are various TCP approaches and CT techniques in the literature, there are still limited studies focusing on both of this techniques used together to improve the performance of the software testing. Thus, this review is to perform a systematic literature review (SLR) on the latest used TCP in CT. The point of this study is to summarise all of the current related research papers based on the research questions that were constructed. The author believes that literature study of this area will grow timely due to the increasing amount of paper published regarding TCP and CT problems.

The remaining of this paper is organised as follow. Section 2 explains the related works and Section 3 is the overview of methodology and research questions. Section 4 describes the result trends and identifies research opportunities. Section 5 is the discussion related to the research questions presented and Section 6 will conclude the SLR study.

2 BACKGROUND AND RELATED WORK

In this section, the discussion will be focusing on the previous studies that are related to TCP and CT techniques. Very few systematic review papers discuss the test case prioritisation technique or combinatorial testing domain. Based on the collected literature review, there are two systematic literature reviews, one systematic mapping, and two survey papers had been selected. Furthermore, in this section, the author will discuss the problems of test case prioritisation and combinatorial testing used to solve the problems.

The Systematic Mapping Study (SMS) by [9], presents a systematic mapping study in the TCP study that covers the year between 2001 until 2011. Based on their works, from 120 papers there are 10 distributions of TCP methods proposed. The proposed methods include coverage-based, distribution-based, human-based, probabilistic, history-based, requirement-based, model-based, cost-aware, and others. These papers also describe the most used dataset, which is from industrial projects. The SMS concludes the basis improvements of TCP in research and suggests the use of public dataset, emphasises further study on model-based and uses well-known evaluation metrics. On other hand, [10], offered a systematic literature review (SLR) for test case prioritisation in regression testing. Their studies cover from year 1999 until 2016. According to their study, from 69 primary studies, the TCP techniques have the potential to be extended for some of improvements. This paper also emphasised that search-based approach is the most common approach compared to the other approaches and each of the approach has its own advantage and limitation. They also suggested that this technique could be improved in terms of the data used and execution process.

Meanwhile for CT, the survey on combinatorial testing by [11] was completed from 90 selected papers related to CT which then discussed namely on the study of model, test case generation, constraints, failure characterisation, improvement for CT, prioritisation of test cases, metrics, and evaluations used. Furthermore, [12], which 15 selected research papers related to the research questions presented to the state-of-the-art in the CT technique. The existing techniques and limitations of existing simplification techniques were also discussed. It aimed to investigate the relevant improvement in the efficiency of CT algorithms for the systematic evaluation of the simplification CT techniques. Lastly, a survey by [13] elaborated the survey on the use of test case prioritisation technique in the combinatorial testing based on the research papers in between 2005 until 2015. The author classified the methods into two categories, which are pure-prioritisation and prio-regeration. Besides, the weight factor assigns, and the main contributions of each research paper were also emphasised. Next, the evaluation metrics and the dataset that commonly used by researchers were also highlighted. However, in this survey, the author did not specifically focus on the real problem that the published paper was trying to solve.

From SLR, it shows that the use of both TCP and CT techniques is unrestricted to possible enhancement in respective technique. However, most of the studies aimed to improve the efficiency of the proposed techniques in detecting the faults in the software system due to the minimum number of SLR and survey related to the use of both of these techniques.

2.1 Test Case Prioritization

The growing amount of test cases to be executed lead to a higher time consumed during software testing [14]. Hence, TCP technique was introduced to solve this problem. In TCP technique, it reorders the test cases from high priority to low priority according to selection criteria, which could detect faults as early as possible. This is one of the techniques to improve the effectiveness of software testing. Besides, [15], also introduced evaluation metric of Average Percentage Detection Fault (APFD) that currently is common evaluation metrics used in most of TCP technique approaches stated by the available SLR [10].

2.2 Combinatorial Testing

Combinatorial testing (CT), in the context of test cases generation is used to highlight the challenge during the software modification, which consists of multiple parameter inputs and interactions [12]. This technique aims to trigger the faults interaction of parameters during software testing with a covering array (CA) test suite generated by some mechanisms. In the mathematical, combinatorial CA is used as the base in CT [16]. CT gives the option for tester to select parameter values and combines this respective value to produce a covering array. Moreover, there are numerous research papers on the CT predominance t-way testing using the 2-way testing or pairwise testing [17–20]. Usually, the CA consists of specific test data where each tuple of the arrays can be represented as a set of parameter values for a specific test case. On the other hand, if the parameter values are varied, the mathematical will be illustrated as Mixed Covering Array (MCA).

3 RESEARCH METHODOLOGY

The study review attempts to perform the concepts of systematic literature review that were proposed by [21]. Based on the guidelines, the process started from defining the research questions, followed by the search string construction, study selection process, quality assessment criteria and lastly data extraction analysis. During the first phase of the SLR, three main research questions constructed and each of main questions will have sub research questions, which functioned to answer the aim of the SLR. Then, the process continued with constructing the suitable search terms to be used in the search process. Next, the selection phase, which considered the inclusion criteria and exclusion criteria for data extractions, took place. Finally, the last phase is the illustration and analysis of the primary studies results in the SLR.

3.1 Construction Research Question

To overview the existing research on the TCP in CT, the authors had categorised the research questions into three primary studies. The first research question consists of the trend of the TCP in CT from recent years. Next, the questions were grouped according to the recent approaches that frequently being used by the researchers. Furthermore, the evaluation metrics were utilised for each of the articles and papers that had been reviewed. The research question can be categorised as below:

RQ1: What is the trend on the use of TCP in CT published studies?

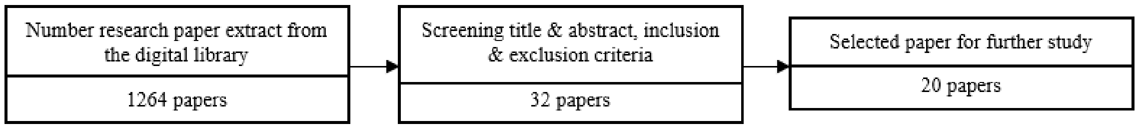

Figure 1: Selection process for selected papers.

RQ1.1: How frequent the related studies discovered for the past few years?

RQ1.2: What is the available TCP approach adapted in published studies?

RQ1.3 What is the available CT strength selected?

RQ2: What is the research topic related to TCP in CT technique?

RQ2.1 How the approaches help solved the related problem in the studies?

RQ3: What is the evaluation metrics used in the published articles and papers?

RQ3.1: How each technique proposed is evaluated?

RQ3.2: What is the dataset used during the evaluation?

3.2 Construction of Search Terms

For the SLR, the authors had constructed a specific search string. The results of the search terms consist of the synonyms for the term "Combinatorial Testing" and "Pairwise-Testing", then include the "Test Case Prioritisation". Therefore, the search string was constructed by adopting the logical expression "OR" and "AND". From the search string, the results show the relevant articles and papers that are related to the subject selected. ("Combinatorial Testing" **OR** "Pairwise Testing") **AND** ("Test Case Prioritisation" **OR** "Test Case Prioritisation")

3.3 Resource Selection

In search of the relevant papers, four online academic paper search engines namely IEEE Xplore, ACM Digital Library, Science Direct, Springer and Google Scholar were utilised. Papers published between 2020 until 2011 were included in this study. The resource repository focused on the related use of TCP and CT techniques, which includes the abovementioned search terms.

3.4 Publication Selection

In summary, the focuses are on English-written studies, papers that are accessible in digital library, and verify each of the articles and papers based on the goals of our mapping studies. Meanwhile, Figure 1 shows the paper selection process, which consist of 1,264 papers using the defined search terms. After screening the titles and abstracts, only 32 papers are related to the research questions that had been constructed. Each of the papers considered the inclusion and exclusion criteria. Thus, 3 inclusion criteria had been used during this process including the paper must be related to TCP and CT, Software Testing and Software Engineering. Then, 4 exclusion criteria include the paper is not written in English, out of scope to TCP and CT, not available online and not technical report and workshop papers. Finally, 20 papers were selected in answering the research question on SLR study.

Table 1: Result of quality score for selected studies

Paper Reference	Q1	Q2	Q3	Score
[22]	1	1	1	3
[23]	1	0.5	1	2.5
[33]	1	1	1	3
[24]	1	1	1	3
[25]	1	1	0.5	2.5
[26]	1	1	0	2
[30]	1	0.5	1	2.5
[31]	1	1	1	3
[32]	1	1	1	3
[34]	1	1	1	3
[35]	1	1	1	3
[36]	1	1	1	3
[37]	1	1	0.5	2.5
[40]	0.5	1	1	2.5
[41]	0.5	1	1	2.5
[28]	1	1	1	3
[27]	1	1	1	3
[29]	1	1	0	2
[38]	1	1	1	3
[39]	1	1	1	3

3.5 Quality Assessment

After the selected process went through the inclusion and exclusion phase, it was chosen according to the quality assessment for evaluation. This is based on 3 quality assessments that had been decided. For the first quality (Q1) is "Does the study stated the aim and objective clearly?". Then, for second quality (Q2) is "Does the study explain the approach or method explicitly?". Lastly, the third quality Q3 is "Did the study evaluate the proposed techniques with reliable evaluation metrics?". From the quality assessment, there are three optional answers for each quality assessment namely "yes" = 1, "partial" = 0.5 and "no" = 0. The selected papers were considered if the assign score is above 2, as stated in Table 1

4 RESULT

This section will discuss on the results with respect to the research questions. On this part, the primary studies were presented first, followed by each of research questions and different answered sub-research questions.

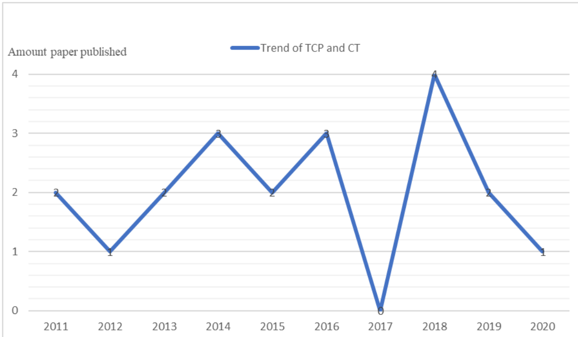

Figure 2: The trend of published paper each year.

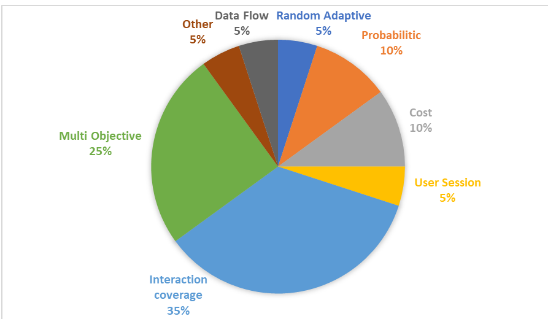

Figure 3: Available approach in selected paper.

RQ1: What is the trend on the use of TCP and CT published?

The aim of this research question is to investigate the trend on the use of TCP in CT published in recent years. The use of TCP in CT is common in the software testing process. The process started with test cases generated using the available CT approach, and test cases were prioritised using the available TCP approach. While for the TCP approach, it is described in details in RQ1.2, followed by the available strength used in CT approach in RQ1.3.

RQ1.1: How frequent the related studies discovered for the past few years?

Based on the collected research papers, the number of studies related to test case prioritisation in combinatorial testing is very limited and uncertain between year 2011 until 2020 which is shown on Figure 2. Since both of the techniques have been proved to have a good fault detection ability for the software under testing (SUT) [9–11], this opens the possibility for researcher to explore more on this specific area of study. Furthermore, for TCP, there is different of approaches available that can be integrated with combinatorial testing.

RQ1.2: What is the available TCP adapted in the published studies?

From the SLR studies, the distribution of TCP in CT techniques was illustrated in Figure 3. The majority of the research papers interaction coverage-based approach were implemented, which contributes to 35 percent while interaction multi-objective is 25 percent. Next, 10 percent of selected papers adapted a cost-based and probabilistic approach. The remaining approaches, which contribute 5 percent of the selected approach, consist of data flow, random adaptive strategy, and user session. Lastly, another 5 percent represents the approach utilised different kind of strategies, such as MC-MIPOG, which are enhancement strategies based on the parameter order strategy (IPOG).

RQ1.3: What is the available CT strength selected?

Based on the analysis, there are 3 available t-way testing or combinatorial testing strengths were used. Firstly, the variable strength [22–29], the combination observes the different levels of strength which could influence the possibility to prompt the fault during the test execution or reduce the test case for software testing. Secondly, fixed strength, [30–36] this strength selection only focuses on specific strength values, for example, 2-way testing (pairwise testing) or 3-way testing which depend on the respective aim of each research paper. Next, the repeated fixed [37, 38], this strength considers the same strength value and repeat the process until the aim of the research can be achieved. From the study, the author observed that the strength of CT is usually in the range between 2 and 6. The higher value of CT strength results in more test cases to be executed. Thus, this could reduce the efficiency of test cases execution.

RQ2: What is the research topic related to TCP in CT technique?

For this research question, the selected primary studies were thoroughly examined concerning how each research paper conducts their study since different papers introduced different problem and procedure on how they conducted the studies. Thus, this section discusses the possible topic that can be prompted from all the selected papers.

RQ2.1: How the approaches help solving the related problem in the studies?

To answer this research question, the author focus on the problem statement and proposed solution that had been suggested in respective research papers. The most recent published paper by [37], concerning the issues of most of the techniques selected appropriate strength values or also called fixed strength value before the software testing. Based on the study, they focus to keep using the small prioritisation strength coverage repeatedly that had not been covered yet during the test case execution. While in previous [38] paper, the author also proposed the same strategy but considering high strength coverage repeatedly to study the effectiveness of different levels of strength. Then, [31] also aims to understand the impact of prioritisation based on the higher strength criteria. On other hand, some of the studies also aimed to reduce the amount of the test cases before prioritising each of executed test cases which could improve the effectiveness in maintaining the testing phase [25, 26, 30]. From the work of [26], the study aimed to gain the minimum number of test cases generated and prioritised the test cases as the output of the study. In this study, they have used the existing t-way strategies, MC-MIPOG, and utilised proposed algorithms to prioritise the test cases.

Furthermore, some of the studies also emphasised the enhancement of the existing approach or proposed different algorithm, and strategy to execute the TCP technique and CT [22, 24, 27–29, 32, 33, 36, 39–41]. Based on [32] study, they adapted the pairwise test data generation to find the optimal test suites and proposed the use of three different evolutionary algorithms (prioritised genetic

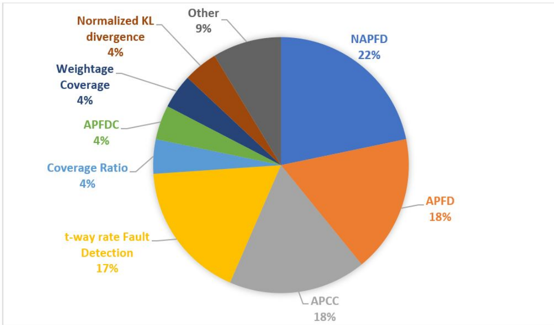

Figure 4: Evaluation metrics used on the studies

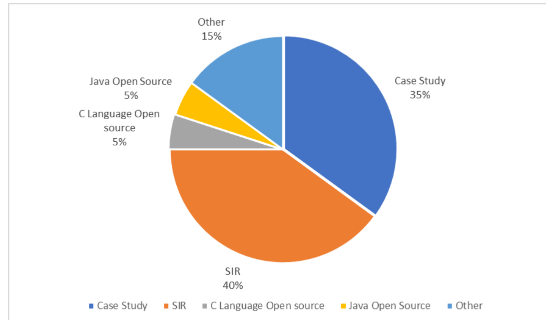

Figure 5: The dataset used to examine the studies

solver, prioritised pairwise combination, plain pairwise sorting) to generate and prioritise the test cases. While, [40], considered the multi-objective to prioritise the test cases selection since most of the study are relying on a single objective, ignoring the possible gain of mixing multiple criteria to seek the faults in the testing process. Next, a few studies also focused on the cost constraint in the study [23, 34, 35]. According to [34], they had considered the cost of reconfiguring parameter values where switching cost of test cases also can influence the orders of the test cases execution. They proposed a greedy algorithm and graph-based algorithm, which resulted in the test suite reduction and helped to detect the fault among the parameter interactions. Moreover, [35], stated that from the previous work of interaction coverage studies, the cost for each of the test cases were often ignored. Thus, they also proposed the algorithms that include the cost of the execution of the study.

RQ3: What is the evaluation metrics used in the published papers?

Based on the primary studies collected, the evaluation metrics used by the studies were examined. Some adopted the existing evaluation metrics from past studies, and several papers introduced different types of evaluation metrics.

RQ3.1: How each approach is evaluated to measure the performance of the proposed technique?

Based on the SLR studies, numerous evaluation metrics have been proposed according to the selected papers illustrated in Figure 4. The majority of the studies used the Normalized Average Percentage Fault Detection (NAPFD) with 23 percent. While for Average Percentage Fault Detection (APFD) and Average Percentage Covering-array (APCC) recorded 18 percent of the common evaluation metrics used in the selected papers. The next common evaluation metric used is t-way rate fault detection (t-RFD) which consists of 17 percent of the selected papers. Another remaining evaluation metrics shared 4 percent from the selected research paper weightage coverage namely Normalized KL divergence, Average Percentage Fault Detection with cost cognizant (APFD*c*) and coverage ratio. Lastly, other metrics contribute to 9 percent.

RQ3.2: What the dataset used during the evaluation?

The process to extract the information based on the dataset that was commonly used during the experimental setup was illustrated in Figure 5. The majority of the selected papers used Software Infrastructure Repository (SIR) including C programming (flex, grep, make, and gzip) and Java programming (siena and nanoxml). 5 percent of the studies used a public dataset including C and Java language. Meanwhile, 25 percent of the study used the case study dataset (e-commerce system, IoT system, Drupal case study, and others) while the other 15 percent used a private dataset. It is clear that most of the researchers selected the SIR dataset since it can be accessed easily and it provides reliable data for the experimental setup.

5 DISCUSSION

From From the literature review, the result of the study shows various proposed techniques that had been widely published in the past years. Firstly, this paper aims to identify the status and how frequent the published papers relate to the use of both this technique in between the year 2011 until 2020 that was illustrated more in RQ1. From performing the analysis, which is answering the RQ1, it shows that the area of study is still in the minimal amount of discussion compared to the papers that specifically focus on TCP and CT independently. Besides, from the goal of the systematic literature review, this paper attempts to identify the main topic that can be improved in this research study. This goal was discussed in detail in RQ2. The major topic discussed in this section is the enhancement of the previous works of the existing researches. Most of the proposed works had taken various types of optimisation metrics into consideration and proposed new strategies and methods to execute this techniques, which included different types of TCP techniques such as the coverage-based, interaction coverage, and cost-based approach. Furthermore, the evaluation metrics and dataset used were thoroughly elaborated in RQ3. The result of the review shows that the NAPFD is commonly used to evaluate the effectiveness of their proposed techniques. From the review, some authors suggest that the use of NAPFD is more reliable since the amount of test cases has been reduced and the amount of fault detection could change in the testing process. Since this metric is suitable to examine the effectiveness of the proposed technique to prompt faults. While for the dataset, SIR repository dataset is the most popular compared to others.

Based on the SLR studies, the author found out that there is more that can be improved. From the study, it was discovered that there are numerous of different approaches in test case prioritisation available apart from what had been listed on the SLR. Thus, more enhancements can be explored in this area of study. Secondly, most of the proposed techniques ignore the constrain factor of cost and severity level of execution for each of the test cases. In real world situation, each of test cases is not similar in terms of cost and severity level. This issue should be emphasised for future work. Lastly, most of the approaches are proposed theoretical and using the standard benchmark dataset to evaluate the effectiveness of the testing. Hence, the author suggests that the software tester expert should examine the approaches from the real scenario.

The frequent threats to validity of the SLR are associated with the imperfect paper collection of the primary studies and the minimal amount of the primary studies that were chosen during this SLR studies. From the selecting process of the primary studies paper, it is clear that the selection of search string is crucial. The problems arise when the selection of search string related to the topic would use different keywords, which makes it possible to be overlooked. Besides, the SLR considered 20 papers out of 1264 papers to be the primary studies after undergoing the quality assessment evaluation.

6 CONCLUSION

The systematic literature review aims to find out the status of research on the use of TCP in CT techniques. The analysis of the SLR is based on the constructed research questions. The research question started with observing the trend of the current publication for related papers in this area of study. The result of the study shows that the trend for 8 different types of TCP approaches used in CT and interaction coverage-based is been identified. The result indicates that the enhancements of previous works on the TCP approaches have become a major topic that is on the top of the discussion. This includes which approach in TCP techniques is suitable to be adapted in the CT that could improve the effectiveness and efficiency of future study. The SLR analysis is a proof of significant contributions for this research study to fill in the gap of different proposed techniques.

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